

Human-Computer Interface Evaluation and Adaptive Control Method Based on Analytic Hierarchy Process

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Abstract

To improve the overall design quality of human-computer interaction interfaces, an interface evaluation and adaptive control scheme based on the Analytic Hierarchy Process (AHP) is introduced. This method meticulously constructs a three-dimensional evaluation index system encompassing practicality, efficiency, and user experience, and designs a multi-level weighting algorithm to complete the comprehensive scoring and ranking of interface schemes. Based on this, a mapping relationship is established between the evaluation results and control parameters, forming a closed-loop control strategy for dynamic interface optimization. Experimental results fully demonstrate that this method can accurately identify the strengths and weaknesses of interface design, providing scientific support for the adaptive adjustment of interface elements.

CCS Concepts

• **Human-centered computing; Human computer interaction (HCI); Interaction paradigms;**

Keywords

Analytic Hierarchy Process (AHP), Human-Computer Interaction (HCI), Interface Evaluation, Adaptive Control, User Experience

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1 Introduction

Human-computer interfaces (HCIs) are the primary channel for communication between users and software systems. Their design directly impacts system usability and user experience. Traditional interface design often relies on the designer's subjective experience, lacking a systematic evaluation standard to objectively compare the merits of different design solutions [1-2]. As software system functions become increasingly complex, scientifically evaluating interface performance and dynamically adjusting it has become a significant technical challenge. The Analytic Hierarchy Process

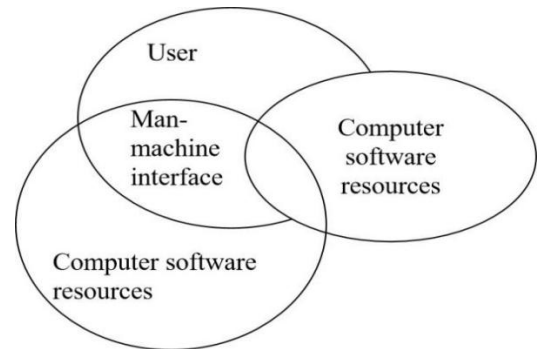


Figure 1: Schematic diagram of human-computer interface.

(AHP), as a mature multi-criteria decision-making method, can decompose complex evaluation problems into multiple levels and determine the weights of each factor through pairwise comparisons. This paper applies AHP to HCI evaluation, establishing a closed-loop system that combines evaluation and control to achieve quantitative evaluation and adaptive improvement of interface design.

2 Hierarchical Evaluation Model of Human-Computer Interface

2.1 Basic Principles of Human-Computer Interaction Interface

Human-computer interaction interfaces are a key bridge for communication between users and software systems, and their design level directly affects the ease of operation and user experience. A refined human-computer interaction interface generally refers to the relevant interface in a computer system, also known as a user interface. It is mainly responsible for the information transmission and interaction between humans and computers, constituting the computer application environment [3-4]. Its basic principle is shown in Figure 1.

2.2 Interface Evaluation Requirements and Problem Analysis

Software interface optimization typically requires iterative adjustments. Before each adjustment, the current solution must be evaluated to clarify the optimization path [5]. In practice, reviewers often offer subjective opinions based on their own experience. Reviewers with different professional backgrounds focus on different aspects, which can lead to substantially different judgments on the same solution. This subjective difference makes it difficult for the

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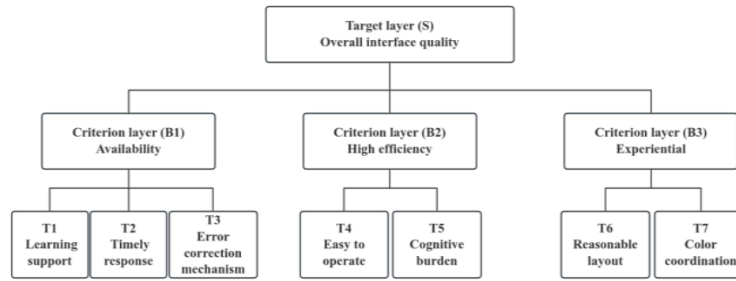


Figure 2: Interface Evaluation Hierarchical Model.

design team to reach a consensus on improvement. Furthermore, the interface effect is determined by multiple factors, each with varying degrees of influence, which traditional evaluation methods struggle to quantify. Among various multi-criteria decision-making methods, AHP has obvious advantages in evaluating the interface. TOPSIS needs complete digital data, and it doesn't perform well when dealing with experts' qualitative opinions [6]. PROMETHEE relies on a preset preference function, and its parameters are difficult to be determined objectively [7]. On the contrary, AHP can change the judges' subjective preferences into measurable values through pairwise comparison, and its hierarchical structure is naturally suitable for the multi-faceted characteristics of interface quality. In addition, its consistency check function can find logical contradictions in expert evaluation, which can improve the reliability of evaluation results.

2.3 Multi-dimensional evaluation criteria framework

The main challenge in interface quality assessment lies in how to decompose the overall design goals into individually measurable components [8]. Based on human-computer interaction theory, the user's interaction with the interface can be divided into three stages: cognition, operation, and perception. The assessment criteria design needs to comprehensively cover these three stages while ensuring their independence. The cognition stage focuses on the user's understanding of the interface functions and learning efficiency; the operation stage focuses on the smoothness and timeliness of task completion; and the perception stage examines the clarity and comfort of visual information. Based on this, this study constructs three independent assessment dimensions and develops specific quantitative indicators for each dimension, forming a hierarchical model as shown in Figure 2. This framework lays a structured foundation for subsequent weight allocation and comprehensive evaluation using the Analytic Hierarchy Process (AHP).

As shown in Figure 2, the criterion layer comprises three levels: usability, efficiency, and user experience. Usability focuses on the completeness of the system's core functions, such as learning support, responsiveness, auxiliary functions, and error correction mechanisms. Efficiency emphasizes ease of operation and task execution efficiency, focusing on reducing the user's cognitive burden. User experience focuses on the visual presentation, including

the scientific layout of the interface, the orderly arrangement of information, and the harmony of color usage. This hierarchical system transforms the abstract concept of interface quality into quantifiable and assessable parameters, establishing a theoretical basis for the systematic evaluation of different design schemes.

3 Implementation of Interface Evaluation Algorithm Based on Ahp

3.1 Evaluation Data Collection and Preprocessing

The data collection methods for interface evaluation are mainly divided into two categories: expert scoring and user feedback. Expert evaluation involves professionals with a background in human-computer interaction design comparing each evaluation indicator pairwise to determine its relative weight. Before the formal evaluation begins, standardized indicator explanation documents need to be distributed to the review experts to ensure that all participants have a unified understanding of the specific meaning of each indicator. User data is collected through questionnaires and usability testing to gather information about users' actual operating experiences. A 1-9 scale is used to construct the judgment matrix, and its transformation relationship is as follows:

$$S_{ij} = \frac{R_i}{R_j} \quad (1)$$

Where S_{ij} represents the importance scale value of factor i relative to factor j , and R_i and R_j are the original score data of factors i and j , respectively. A scale value of 1 represents equal importance, while 3, 5, 7, and 9 represent slightly important, relatively important, highly important, and absolutely important, respectively. To ensure the accuracy of the evaluation conclusions, consistency checks must be performed on the collected data from multiple experts to eliminate outliers with obvious logical conflicts. The individual judgment matrices of 15 experts all passed the test, and the final combination matrix was obtained by calculating the geometric average of their respective elements.

3.2 Multi-level weight calculation algorithm

Weight calculation, a core step in the analytic hierarchy process (AHP), aims to determine the relative importance of each factor through a judgment matrix. This paper chooses the square root method to calculate the weight vector. This method is simple to

Table 1: Calculation results of the human-computer interface evaluation system S.

Solution t	The weight of T to B			
	B1	B2	B3	Overall weighting (T and B)
T ₁	0.0998	0.2969	0.2369	0.0168
T ₂	0.3726	0.2896	0.1259	0.1469
T ₃	0.1859	0.4569	0.4589	0.1325
T ₄	0.3569	0.7892	0.2698	0.4596
Solution t	The weight of T to B			
T ₅	0.2369	0.4263	0.7856	0.4896
T ₆	0.2398	0.1259	0.5983	0.8963
T ₇	0.1297	0.6392	0.4593	0.1589
λ	4.0105	4.0406	4.0896	0.1789
CL	0.0036	0.0125	0.0293	0.3694
CR	0.038	0.0125	0.0325	0.2593

operate and yields reliable results, making it suitable for determining the weights of small to medium-sized judgment matrices. The calculation formula is as follows:

$$W_i = \frac{\sqrt[n]{\prod_{j=1}^n a_{ij}}}{\sum_{k=1}^n \sqrt[n]{\prod_{j=1}^n a_{kj}}} \quad (2)$$

Where W_i is the weight value of the i factor, a_{ij} is the element in the i row and j column of the judgment matrix, and n is the total number of factors. The formula calculates the geometric mean of each row in the judgment matrix and then normalizes it to obtain the weight vector. To ensure the scientific validity of the judgment matrix, a consistency test is required, and the consistency ratio CR is calculated. The test is considered successful when $CR < 0.1$. For the evaluation index system established in this study, the weights of the three criteria (usefulness B1, efficiency B2, and comfort B3) and the seven evaluation factors (T1 to T7) are calculated sequentially. The specific results of the weight calculation values, the maximum eigenvalue λ , the consistency index CI, and the consistency ratio CR for each level are detailed in Table 1.

3.3 Overall Scoring and Ranking of Interface Schemes

After obtaining the weights at each level, a weighted summation model is used to calculate the comprehensive evaluation value of the interface design scheme [9]. For any specific design scheme, its specific score on each evaluation indicator is first determined. This score is obtained through user testing results or expert scoring and uniformly converted to a standardized range of 0-1. Then, a weighted calculation is performed based on the weight coefficients of each evaluation indicator to obtain the score of the scheme in each criterion dimension. Finally, the criterion layer scores are combined with their weights and weighted again to obtain the final comprehensive score of the scheme [10]. To facilitate comparative analysis of various schemes, the evaluation results of all candidate schemes are summarized into a hierarchical ranking table. For three different interface design schemes F1, F2, and F3, testing is conducted on seven evaluation indicators from T1 to T7. Based on the weight values assigned in Table 1, the comprehensive ranking

results of each scheme are finally obtained by using the analytic hierarchy process (AHP) (see Table 2 for details). The comparison between comprehensive scores can clearly reflect the differences in the advantages and disadvantages of different schemes, while the distribution analysis of scores in each dimension can reveal the outstanding advantages and areas for improvement of each scheme.

3.4 Evaluation Process and Algorithm Framework

To meet the needs of practical engineering, this paper summarizes relevant technical methods into a unified evaluation process. First, the project team selects interface solutions to be evaluated and records the version number and core differences of each solution. Then, an evaluation briefing is held to explain the meaning of each indicator and scoring criteria to the evaluation experts, ensuring a consensus on the evaluation scale. After experts independently complete their scoring, the algorithm module automatically summarizes the scoring results and identifies outliers, marking and reviewing scores that significantly deviate from the average level. After weight allocation, the system generates an evaluation report, which includes the total score for each solution, the score distribution for each criterion, and weak points requiring key improvement. The report uses charts to present the core conclusions, helping the design team efficiently identify optimization directions. The entire evaluation process, from information collection to report output, takes approximately 3 to 5 working days, with the expert scoring stage taking the most time, accounting for about 60% of the entire cycle.

4 Instance Verification and Method Analysis

4.1 Interface Scheme Evaluation Experiment

To verify the practical effectiveness of the proposed evaluation method, a comparative test was conducted on three interface designs of an enterprise-level management system. The research involved 15 experts (including UX designers, software engineers and usability researchers, each with at least 3 years of professional background) and 30 real end-users. Solution F1, using a classic menu layout, emphasizes comprehensive functionality and system

Table 2: Overall Hierarchical Ranking.

	T ₁	T ₂	T ₃	T ₄	T ₅	T ₆	T ₇	Overall score
	0.0162	0.1437	0.0301	0.3237	0.0975	0.0597	0.3288	
F ₁	0.6585	0.5905	0.5498	0.0926	0.1261	0.2159	0.1638	0.2212
F ₂	0.1851	0.2765	0.2098	0.2923	0.4162	0.6819	0.5395	0.4024
F ₃	0.1563	0.1282	0.2403	0.6154	0.4578	0.1026	0.2968	0.3756

Table 3: Comparison of User Test Data for Three Interface Schemes.

Test metrics	Option F1	Option F2	Option F3
Average task time (s)	118.5	82.3	96.7
Frequency of operational errors (times/task)	1.4	1.1	1.8
Subjective satisfaction rating	3.56	4.08	4.21

reliability, with well-considered user learning and error recovery mechanisms. Solution F2 focuses on improving operational efficiency by providing quick access areas and intelligent guidance to help users reduce repetitive tasks. Solution F3 prioritizes visual experience optimization, employing a modern flat design concept and a dynamic color scheme to ensure clear and logical presentation of interface information. During the study, each participant was required to complete three main tasks for each solution: finding documents, submitting forms, and managing the approval process. The researchers recorded task completion time and error frequency, and used a 5-point Likert scale to measure user satisfaction. Simultaneously, evaluation experts compared and scored each solution in pairs according to established standards. Specific test results are shown in Table 3.

As can be seen from the data in Table 3, Solution F2 performs well in both task completion time and error frequency, demonstrating high operational efficiency. Solution F3 received the highest subjective satisfaction rating, which confirms its superiority in visual design. However, the high error frequency also exposes its limitations in practical use. Although Solution F1 takes longer to complete a task, it has fewer errors than Solution F3, indicating that its functional architecture is reasonable and its error correction design is effective. This provides a clear direction for subsequent interface optimization.

4.2 Evaluation-based Interface Adaptive Control

The evaluation results prove that the rule-based method can be used to improve the interface design. Specifically, if the efficiency index is lower than 0.6 and the task time is 20% longer than the standard, then the commonly used functions are moved to the main screen; If the usability score is lower than 0.5 and each task makes more than 1.5 mistakes, then the function of preventing mistakes should be strengthened; If the user experience score is lower than 0.7, then the visual design should be optimized. After applying these rules, Solution F1 reached the efficiency standard, so we implemented an adaptive adjustment system. Firstly, adjustments were made to the interface structure, moving frequently used functional modules from secondary menus to the main page to simplify operation. Analysis of user behavior data revealed that document retrieval,

data export, and approval processes were the three most frequently used functions, which were then set as shortcuts. Secondly, improvements were made to the visual presentation by reducing information density and optimizing the color scheme. The interactive experience was also enhanced by improving real-time operation feedback. User testing was conducted on the improved interface solution using these optimization measures. Data showed that average task completion time decreased, the error rate significantly decreased, and user satisfaction scores also improved. This method confirms that targeted optimization based on evaluation results is reasonable and efficient.

4.3 Analysis of the applicability and limitations of the method

The method described in this paper has several limitations in practical applications. In terms of interface type, this method is primarily suited for functional business system interfaces. For game interfaces emphasizing immersive experiences or monitoring interfaces focusing on data-intensive presentation, the comprehensiveness of existing evaluation metrics may be insufficient. Regarding the evaluation timeline, it is recommended to choose a time when the prototype design is completed but before formal development begins, as the cost of adjusting the solution is lower and the differences between different solutions are more significant at this stage. In terms of data collection, expert opinions are currently the primary basis, with actual user behavior data serving only as supplementary reference. When there are significant differences in the characteristics of the product's user group, it may be necessary to set differentiated weighting coefficients for different user groups. Furthermore, the evaluation results only reflect the relative ranking between different solutions, not the absolute quality level of the interface; therefore, the overall score should not be used as the sole criterion for judging whether an interface is acceptable.

5 Conclusion

This paper establishes a human-computer interface (HCI) evaluation and control framework based on the Analytic Hierarchy Process (AHP). The proposed multi-level weight calculation algorithm can

automatically process evaluation data, thereby improving the efficiency and objectivity of the evaluation process. Example tests demonstrate that this method can accurately identify the performance differences between different interface solutions, providing data-driven support for design decisions. Furthermore, the constructed evaluation-control mapping mechanism enables the interface to adaptively adjust based on the evaluation results. Future research can be conducted in the following areas: first, improving the evaluation index system to adapt to the evaluation requirements of special scenarios such as game interfaces and data visualization; second, utilizing user behavior logs to construct a weight allocation mechanism for different groups; and third, promoting the deep integration of evaluation work and development processes to accelerate the transformation from evaluation feedback to design optimization.

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